



Homework 2

Regression · Quarto Blogging

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Directions

- Submit your **Quarto document** for **Part 1** to Brightspace using this filename format:
 - `dan1-320-hw2-LASTNAME-FIRSTNAME.qmd`
(e.g., `dan1-320-hw2-choe-byeonghak.qmd`)
- **Due:** March 2, 2026, 3:30 PM (ET)
- Questions? Email Byeong-Hak bchoe@geneseo.edu.

Required R Packages

```
library(tidyverse)
library(broom)
library(stargazer)
library(rmarkdown)
library(margins)
library(yardstick)
library(wvPlots)
library(pROC)
```

Part 1. Regression



Consider the `homes` DataFrame from the 2004 American Housing Survey, which includes data on home values, demographics, schools, income, finance, mortgages, sales, neighborhood characteristics, noise, smells, state geography, and urban classification.

```
homes <- read_csv(
  'https://bcdanl.github.io/data/american_housing_survey.csv'
)

homes <- homes |>
  mutate(GT20DWN = ifelse((LPRICE - AMMORT) / LPRICE > 0.2,
                           1, 0))
```

Show entries

Search:

	LPRICE	VALUE	STATE	METRO	ZINC2	HHGRAD	BATHS	BEDRMS
1	85000	150000	GA	rural	15600	No HS	2	3
2	76500	130000	GA	rural	61001	HS Grad	2	3
3	93900	135000	GA	rural	38700	HS Grad	2	3
4	100000	140000	GA	rural	80000	No HS	3	4
5	100000	135000	GA	rural	61000	HS Grad	2	3
6	96000	125000	GA	rural	265000	Bach	2	3

	LPRICE ▴ ▾	VALUE ▴ ▾	STATE ▴ ▾	METRO ▴ ▾	ZINC2 ▴ ▾	HHGRAD ▴ ▾	BATHS ▴ ▾	BEDRMS ▴ ▾
7	130500	135000	GA	rural	42000	Bach	2	3
8	120000	130000	GA	rural	87000	HS Grad	2	4
9	199900	250000	GA	rural	87800	No HS	3	3
10	105000	110000	GA	rural	27030	No HS	2	3

Showing 1 to 10 of 15,565 entries

Previous

1

2

3

4

5

...

1,557

Next

Variable Description

Variable	Description
LPRICE	Purchase price of unit and land
VALUE	Current market value of unit
STATE	State code
METRO	Central city/suburban status
ZINC2	Household income
HHGRAD	Educational level of householder
BATHS	Number of full bathrooms in unit
BEDRMS	Number of bedrooms in unit
PER	Number of persons in household
ZADULT	Number of adults (18+) in household
NUNITS	Number of units in building
EAPTBL	Apartment buildings within 1/2 block of unit
ECOM1	Business/institutions within 1/2 block
ECOM2	Factories/other industry within 1/2 block
EGREEN	Open spaces within 1/2 block of unit

Variable	Description
EJUNK	Trash/junk in streets/properties in 1/2 block
ELOW1	Single-family town/rowhouses in 1/2 block
ESFD	Single-family homes within 1/2 block
ETRANS	RR/airport/4-lane highway within 1/2 block
EABAN	Abandoned/vandalized buildings within 1/2 block
HOWH	Rating of unit as a place to live
HOWN	Rating of neighborhood as a place to live
ODORA	Neighborhood has bad smells
STRNA	Neighborhood has heavy street noise/traffic
FRSTHO	First home
AMMORT	Amount of 1st mortgage when acquired
INTW	Interest rate of 1st mortgage (whole number %)
MATBUY	Got 1st mortgage in the same year bought unit
DWNPAY	Main source of down payment on unit

Question 1

Plot some relationships and tell a story.

Show answer

Question 2

- Fit a **linear regression model** with the following specifications:
 - Outcome variable:** $\log(\text{VALUE})$
 - Predictors:** all but `AMMORT` and `LPRICE`

[Show answer](#)

Question 3

- Refit the **linear regression model**, retaining only statistically significant predictors from Question 2.
- Compare the revised model to the initial model from **Question 2** using:
 - β estimates
 - R^2
 - RMSE
 - Residual plots

[Show answer](#)

Question 4

- Fit a **logistic regression model** with the following specifications:
 - **Outcome variable:** GT20DWN (indicating whether the buyer made a down payment of 20% or more)
 - **Predictors:** All available variables except **AMMORT** and **LPRICE**
- The outcome variable is defined as:

$$\text{GT20DWN} = \begin{cases} 1 & \text{if } \frac{\text{LPRICE} - \text{AMMORT}}{\text{LPRICE}} > 0.2 \\ 0 & \text{otherwise} \end{cases}$$

- Analyze and interpret the following relationships:
- The association between **first-time homeownership** (FRSTHO) and the probability of making a 20%+ down payment.
- The association between **number of bedrooms** (BEDRMS) and the probability of making a 20%+ down payment.

[Show answer](#)

Question 5

- Refit the **logistic regression model**, adding interaction terms:
 - **Predictors**: all previously included predictors in Question 4 **plus** the interaction between **FRSTHO** and **BEDRMS**
- Interpret how the relationship between **BEDRMS** and the probability of a 20%+ down payment varies depending on whether the buyer is a first-time homeowner (**FRSTHO**).

Show answer

Question 6

- Fit separate **logistic regression models** (with the same model specification as in Question 4) for two subsets of home data:
 1. Homes worth $\text{VALUE} \geq 175k$.
 2. Homes worth $\text{VALUE} < 175k$.
- Compare residual deviance, *RMSE*, and classification performance between the two models.

Show answer

Part 2 Task — Write a blog post (Quarto Document)

Write a blog post about **Part 1 of Homework 2 - Beer Markets**, and add it to your online blog.

Requirements (minimum):

1. Create a post using a **Quarto Document** (`.qmd`) and ensure it appears on your Quarto website blog.
2. Include:
 - a short introduction (what the dataset is and what you'll explore)
 - **descriptive statistics** (at least 3 summary numbers)
 - **counting** (e.g., top flavors)
 - **filtering** (at least one meaningful filter)
 - **grouped summaries** (at least one `group_by()` + `summarise()`)
 - **regression models** in Homework 1
3. Optional but encouraged:
 - at least one ggplot visualization

Deliverables:

- The blog post (in your website repo) renders successfully
- The post is accessible from your website's blog listing page
- In your Brightspace submission (Part 1 QMD), include the **URL** to your published blog post



End of Homework 2

powered with github, quarto, and rstudio
byeong-hak choe, 2026